TH series
Industrial Robot

INSTRUCTION MANUAL

CLEAN SPECIFICATIONS

Notice

1. Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
2. Before installing and operating the industrial robot, read through and completely understand this manual.
3. After reading through this manual, keep it nearby for future reference.

Rev.1
April 1, 2020

Shibaura Machine

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Preface

SHIBAURA MACHINE's TH–CR series system robot is designed and manufactured as per the clean specifications, taking over the high-speed and high-precision performance characteristics of the standard TH series robot.

This manual describes the difference in structure, wiring, piping and maintenance of the basic robot from the standard specifications. Before actually unpacking, you are requested to look through this manual.
Also, before using this manual, please understand the name and function of each part, referring to the Transportation and Installation Manual, and Maintenance Manual.

This manual consists of the following three (3) sections.

Section 1 Clean Robot Basic specifications
This section describes the basic specifications, outer dimensions and working envelope of the robot.

Section 2 Dust-Preventive Measures
This section stipulates the dust-preventive measures when using the robot of the clean specifications.

Section 3 Maintenance and Inspection
This section describes the difference in maintenance and inspection of the basic robot from the standard specifications.

Cautions on Safety

This manual contains the important information on the robot and controller to prevent injury to the operators and persons nearby, to prevent damage to assets and to assure correct use.
Make sure that the following details (indications and symbols) are well understood before reading this manual. Always observe the information that is noted.

[Explanation of indications]

Indication	Meaning of indication
 **DANGER**	This means that "incorrect handling will lead to fatalities or serious injuries."
 **CAUTION**	This means that "incorrect handling may lead to personal injuries *1) or physical damage *2)."

*1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long-term medical treatment.

*2) Physical damage refers to damages due to destruction of assets or resources.

[Explanation of symbols]

Symbol	Meaning of symbol
	This means that the action is prohibited (must not be done). Details of the actions actually prohibited are indicated with pictures or words in or near the symbol.
	This means that the action is mandatory (must be done). Details of the actions that must be done are indicated with pictures or words in or near the symbol.
	This means danger. Details of the actual danger are indicated with pictures or words in or near the symbol.
	This means caution. The details of the actual caution are indicated with pictures or words in or near the symbol.

CAUTION

- Be sure to read through the safety instruction manual provided separately before starting actual work to assure the safe work covering from the robot installation to operation.

[Handling the clean robot]

Be sure to observe the following items to use the robot safely.

CAUTION	
 Prohibited	• The user must NEVER replace or remodel parts other than those described in the instruction manual. Otherwise, the robot performance may deteriorate, or faults or accidents will be caused.
 Mandatory	• As the type of grease used for the ball screw spline differs from the standard specifications, be sure to use the grease designated by SHIBAURA MACHINE. Otherwise, the robot performance may deteriorate, or faults or accidents will be caused. • For part replacement, be sure to use the spare parts designated by SHIBAURA MACHINE. • Carry out the maintenance and inspection regularly. Unless they are executed, faults or accidents will be caused.

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Section 1 Clean Robot Basic Specifications

1.1 Basic Robot Specifications

Table 1.1 shows the basic specifications of the clean robot.

Table 1.1 Basic robot specifications

Model		TH250–CR	TH350–CR
Arm length	Full length	250 mm	350 mm
	Arm 1	125 mm	225 mm
	Arm 2	125 mm	125 mm
Working envelope	Axis 1	±115°	±115°
	Axis 2	±140°	±145°
	Axis 3 (vertical)	120 mm	120 mm
	Axis 4 (wrist rotation)	±360°	±360°
Maximum speed	Axis 1	480°/s	300°/s
	Axis 2	480°/s	480°/s
	Axis 3 (vertical)	1067 mm/s	1067 mm/s
	Axis 4 (wrist rotation)	1143°/s	1143°/s
	Composite	3.14 m/s	2.88 m/s
Load	Mass of workpiece to be carried	Max. 3 kg (Load gravity center offset, 70 or less)	Max. 3 kg (Load gravity center offset, 70 or less)
	Permissible inertia moment at the tip	0.017 kgf·m^2	0.017 kgf·m^2
Repeatability	X, Y (plane)	±0.01 mm	±0.01 mm
	Axis 3 (vertical)	±0.01 mm	±0.01 mm
	Axis 4 (wrist rotation)	±0.03°	±0.03°
Tool wiring and air piping		14 cables and 4 pipelines	
Degree of cleanliness		Designed for cleanliness 10 (*Note 1)	
Drive system		AC servo motor	
Mass of the basic body		15 kg	16 kg
Robot controller		TS1000	

Note 1: Cleanliness 10:
When this robot is operated in the clean bench (cleanliness 1 or less) installed in the downflow clean room, the number of fine particles of 0.3 μm or over contained in 1 ft^3 as measured in the plane of the lower end position of axis 3 stroke is ten (10) or less.
Air speed of downflow is 0.3 ~ 0.5 m/s.

1.2 External View and Working Envelope

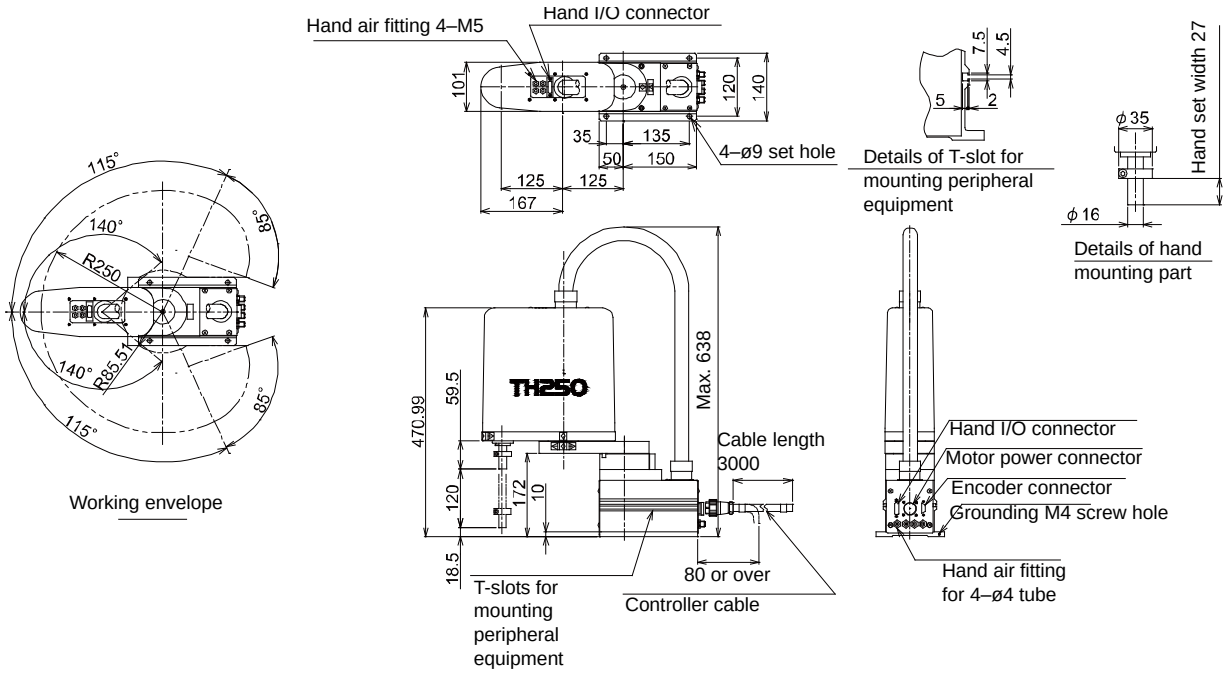


Figure 1.1 External view of TH250–CR

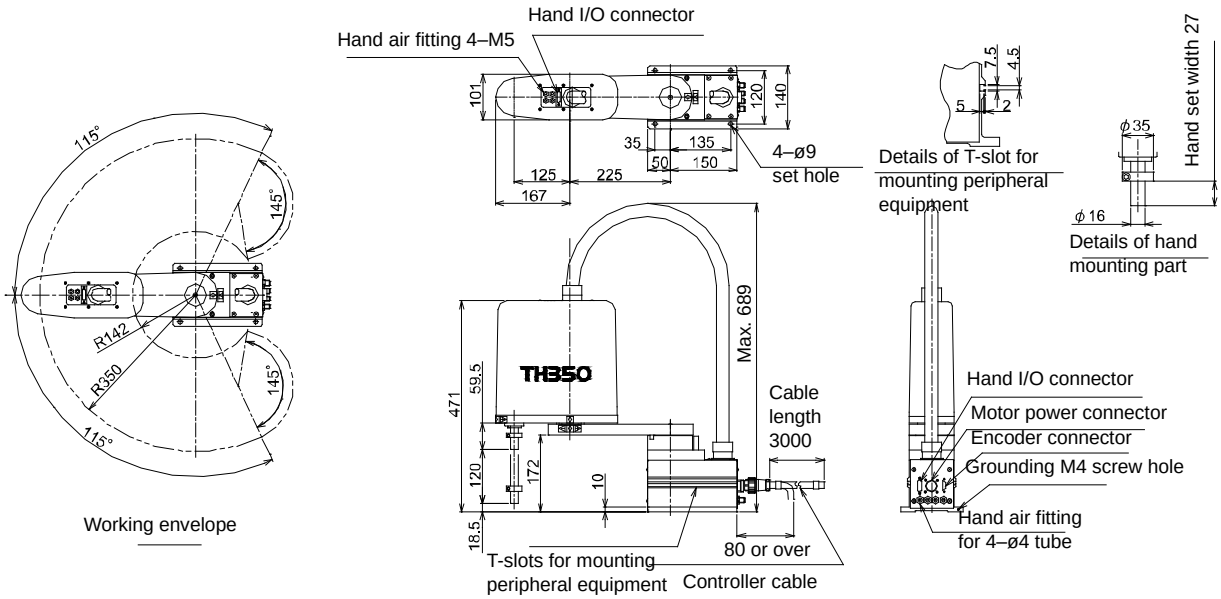


Figure 1.2 External view of TH350–CR

Section 2 Dust-Preventive Measures

2.1 Dust Preventive Measures

1) Ball screw spline
   The ball screw spline shaft is hard chrome plated.
   Balls of the ball screw nut and spline nut are made of stainless steel.
   A dog attached to the ball screw spline shaft is made of stainless steel.
   Clean fluorine-base grease is used.

2) Painting
   The arm 2 cover is coated with electrostatic preventive agent.

3) Bolt and air fitting
   All bolts exposed are made of stainless steel.
   All air fittings exposed are non-electrolytic nickle plated.

4) Totally enclosed structure
   The arm 2 cover is completely sealed with packing set between the bottom of the cover and arm 2 to prevent dust flow into the outside and to increase negative pressure.

5) Robot internal scavenging
   The arm 2 cover interior and axis 4 rotating unit are scavenged by means of an air tube.

6) User's wiring and piping
   The user's wiring and piping are connected on top of the arm 2 cover.
   The user's wiring can be easily removed and attached by means of a connector.

7) Assembly and test
   Assembly and test are conducted in the clean booth of class 10000.

8) Packing
   After cleaning the robot exterior with alcohol, the entire robot is packed in a polyethylene bag for delivery.

As no clean measures are taken on controller TS1000, take utmost care of the place where it is to be installed.

2.2 Scavenging Piping in the Robot

Air tubes are incorporated in the clean robot for scavenging.
Figures 2 show the piping of the air tube.
When operating the robot as the clean robot, scavenge through the air fittings, using a scavenging vacuum pump (or equivalent to be provided by the user). Yardstick for the scavenging volume is 50 liters/min.
If the robot is operated without scavenging, the specified degree of cleanliness cannot be maintained.

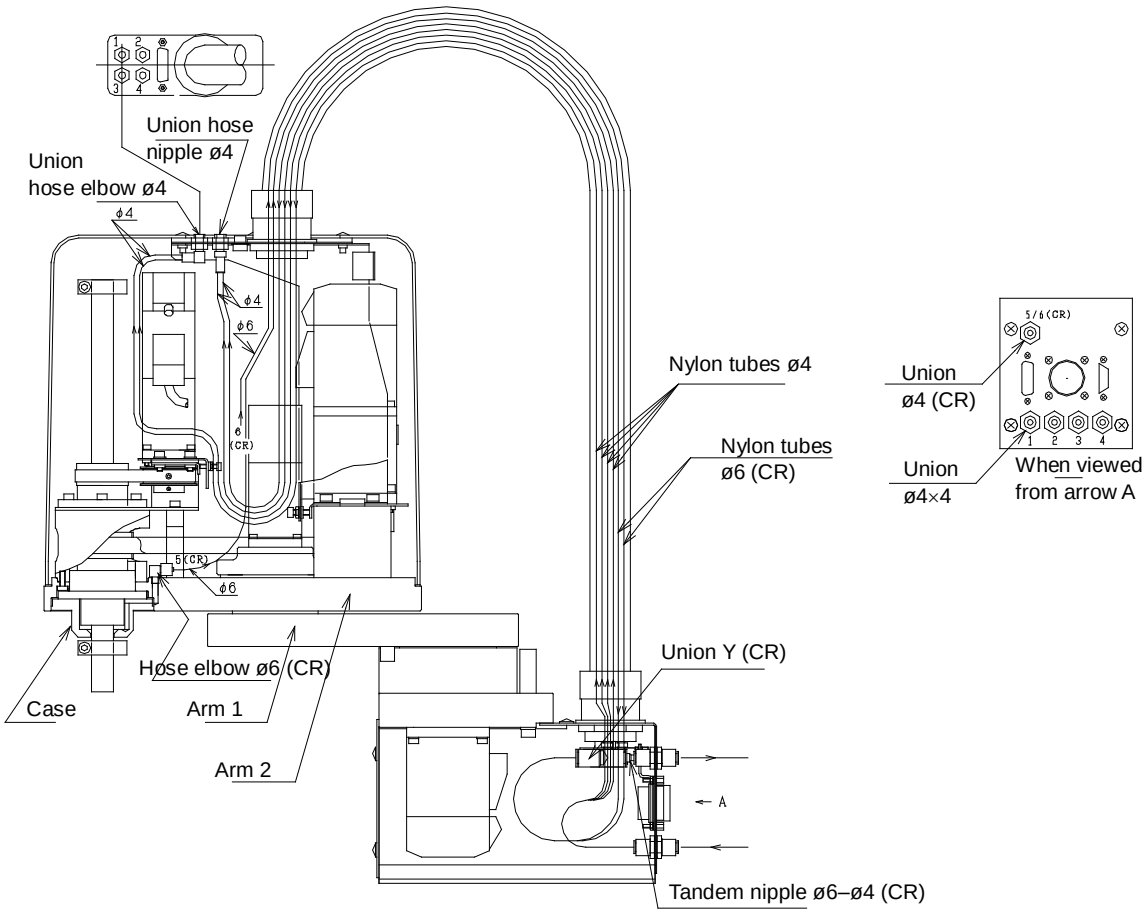


Figure 2 Air piping for scavenging

2.3 Sealing Robot Cover

To maintain a better effect, the clean robot cover is filled up with packing using sealant to eliminate any clearance and to realize a totally closed robot structure. Dust in the robot will not flow into the outside.
Except for the internal inspection and adjustment, DO NOT remove the cover.

Section 3 Maintenance and Inspection

3.1 Maintenance and Inspection Items

The maintenance and inspection schedule and inspection items are the same as in the standard specifications. For details, see Section 1 of the Maintenance Manual. Bear in mind that the type of grease for the ball screw spline differs from the standard specifications.

3.2 Lubricating Tool Shaft and Tool Shaft Support Bearing

For the tool shaft, use the following grease.

- FOMBLIN Grease OT–20 (maker: Audimond)

For the greasing, see Para. 2.3.4 of the Maintenance Manual (STE 71287).

! CAUTION

- Be sure to select the proper grease. If the wrong grease is selected, the cleanliness cannot be maintained and the robot performance will deteriorate, resulting in malfunction.

3.3 Replacement Parts List for Maintenance of Robot Mechanism

Table 3.1 [1] TH250–CR / [2] TH350–CR replacement parts list for maintenance

Part name	Model	Dwg. No./Type	Maker	Q'ty	Remarks
AC servo motor	[1] [2]	MSMA022C2U1	SHIBAURA MACHINE	1	For axis 1
	[1] [2]	MSMA012C2U1		1	For axis 2
	[1] [2]	MSMA012C2R1		1	For axis 3
	[1] [2]	MQMA012C2D1		1	For axis 4
Harmonic drive	[1]	M074824		1	For axis 1
	[2]	M086369		1	For axis 1
	[1] [2]	M074825		1	For axis 2
Lithium battery	[1] [2]	ER6V–C		1	
Planetary reduction gear	[1] [2]	M074826		1	For axis 4
Timing belt	[1] [2]	216–3GT–9		1	For axis 3
	[1] [2]	510–3GT–12		1	For axis 4
Timing pulley	[1] [2]	S700133		1	For axis 3
	[1] [2]	S700132		1	For axis 3
	[1] [2]	S700279		1	For axis 4
	[1] [2]	S700131		1	For axis 4
Ball screw spline	[1] [2]	BNS1616AEFM +300LC5F		1	Clean spec.
Harness	[1]	E99176G02		1	
	[2]	H02972G02		1	
Grease	[1] [2]	4B No. 2	Harmonic Drive Systems	500 g, 2.5 kg	
	[1] [2]	OT–20	Audimond	100 g	

! CAUTION

- All items other than the grease listed above are options. When purchasing them, place an order with us.
- NEVER replace the parts other than the lithium battery by the customer.